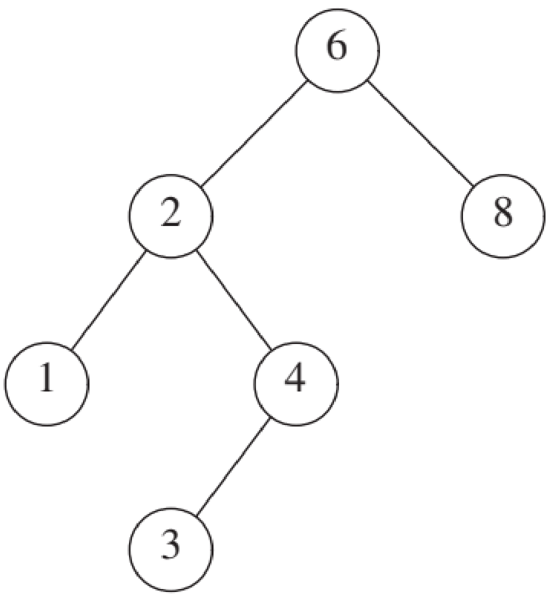
Example for Binary Search Tree: Insert

For example 5.1: if there is the exist BinarySearchTree's instance variable name bst contains the numbers [1, 2, 3, 4, 6, 8] (In-order traversal) as the image following:



Write the steps for the _insert_recursive() function when insert 5 into the instance variable bst. In each round,consists of

1) what are the values of the parameters value and current_node of the _insert_recursive() function, and

2) what statements are executed under the if-else condition that is checked as true.

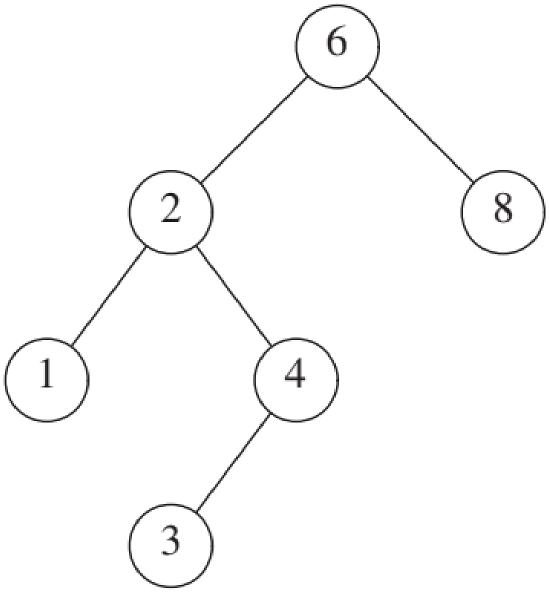
Each iteration must write 2 things.

in main program

bst = BinarySearchTree()

...

bst.insert(5)



Answer:

Each iteration for the _insert_recursive() function when insert 5

First round value = 5 , current_node = 6

_insert_recursive(current_node.left, value)

round 2: value = 5 , current_node =

round 3: value = 5 , current_node =

2 4

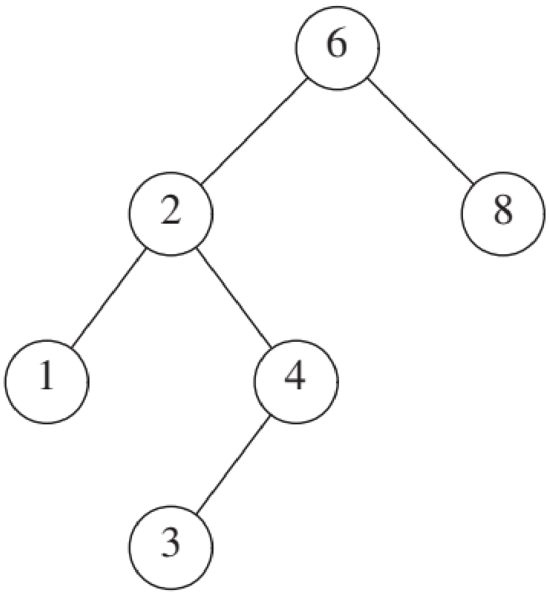
round 4: value = 5 , current_node = **None**

if current_node.right is None: **is True**

current_node.right = Node(value)

Example for Binary Search Tree: Remove (4), one child.

For example 5.2: if there is the exist BinarySearchTree's instance variable name bst contains the numbers [1, 2, 3, 4, 6, 8] (In-order traversal) as the image following:



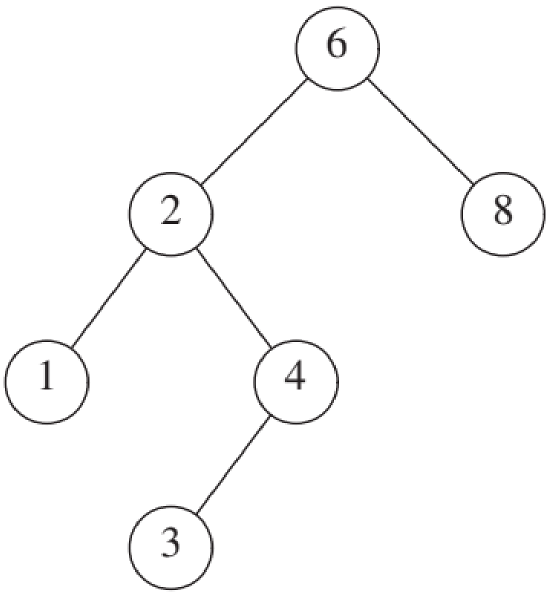
Write the steps for the _delete_recursive() function when remove 4 from the instance variable bst. In each round,consists of

1) what are the values of the parameters value and current_node of the _delete_recursive() function, and

2) what statements are executed under the if-else condition that is checked as true.

Each iteration must write 2 things.

Answer:



Each iteration for the _delete_recursive() function when remove 4

First round value = 4 , current_node = 6

current_node.left = _delete_recursive(current_node.left, value)

round 2: value = 4 , current_node =

current_node.right =

round 3: value = 4 , current_node =

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    elif current_node.right is None:   is True
        return current_node.left
```

2 4